

Predicting LEGO Set Resale Prices using Regression Models

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Abstract

We present a machine learning approach to predicting LEGO set resale prices using structured product metadata. From an initial dataset of over 14,000 sets, we curated 3,607 complete records featuring attributes such as release year, theme, piece count, MSRP, and current resale price. After preprocessing and correlation analysis, we trained and evaluated five regression models – Linear Regression, Decision Tree, Random Forest, Gradient Boosting, and Support Vector Regression – alongside several baselines. Using hold-out validation and hyperparameter tuning, the Decision Tree Regressor (depth = 7) achieved the best results ($R^2 = 0.8011$, RMSE = 59.03). Results show that resale prices are highly predictable from structured features, with tree-based models capturing nonlinear patterns most effectively. We discuss future work involving external data sources, model interpretability, and extensions to other collectible markets.

1 Introduction

Our study focuses on developing a regression model to predict the resale price of LEGO sets based on features such as release year, theme, number of pieces, and original retail price. We use structured tabular data from Kaggle, originally sourced from Brickset.com and Bricklink.com, to train and evaluate the model. Our objective is to provide a data-driven estimation tool that can forecast LEGO resale values with high accuracy. To achieve this, we preprocess the dataset by removing incomplete data – specifically, data points with missing ‘USD_MSRP’ values – and selecting the most relevant features for model training. We evaluate different regression models to identify the best-performing approach for price prediction.

LEGO is the largest toy manufacturer in the world, and in recent years, there has been a growing market for LEGO set investments. Some studies suggest that investing in LEGO can even outperform the SP 500, as certain rare or discontinued sets significantly increase in value over time (Kaggle, 2023). For example, a single Boba Fett minifigure is now valued at over \$3,000, demonstrating the strong appreciation potential within the LEGO secondary market. To analyze this trend, we utilize a dataset containing the most up-to-date list of LEGO sets, along with their features, original list prices, and latest resale prices (Kaggle, 2023).

The LEGO secondary market has gained significant attention as an alternative investment class, with studies suggesting that some sets appreciate in value faster than traditional financial assets like stocks or gold (Cole, 2024). Unlike standard consumer goods, discontinued or rare LEGO sets often experience substantial price increases, making them attractive to collectors and investors alike. According to Professor Victoria Dobrynskaya, an associate professor at the Faculty of Economic Sciences at HSE University in Russia, LEGO sets have demonstrated an average annual return of 10-11%, outperforming many conventional investment strategies such as stocks, bonds, and gold. Some rare sets have even increased in value by over 600% of their original price, driven by factors such as scarcity, franchise popularity, and nostalgia (Dobrynskaya, 2021). This strong resale market highlights the need for a data-driven pricing model that can accurately predict LEGO set values over time.

Despite the growing interest in LEGO investments, price estimation remains largely sub-

jective, relying on expert opinions or manual tracking of resale values across platforms like Bricklink, eBay, and private collector forums (Cole, 2024). This process is time-consuming, inconsistent, and prone to human bias, making it difficult for collectors, resellers, and investors to make well-informed decisions. Additionally, LEGO resale prices are influenced by various factors – including theme, exclusivity, original retail price, and historical demand trends – which makes predicting future values complex. A machine learning-based approach provides a scalable and objective solution by leveraging historical data to uncover pricing patterns and generate reliable resale estimates.

Developing a data-driven LEGO price prediction model benefits multiple stakeholders. Collectors can use it to assess which sets are likely to appreciate in value, guiding their purchasing decisions. Resellers can optimize their inventory by identifying undervalued sets with strong growth potential. Investors can incorporate LEGO sets into their alternative asset portfolios with data-backed insights. Finally, manufacturers and retailers can analyze pricing trends to refine their product strategies and forecast market demand. By combining historical data with advanced regression models, our study aims to bridge the gap between manual pricing strategies and automated predictive analytics, offering a valuable tool for the expanding LEGO investment community.

The dataset used in this study is titled “LEGO Sets and Prices Over Time,” compiled from Brickset.com and Bricklink.com APIs and made publicly available on Kaggle (Kaggle, 2023). The original dataset contains 14,936 LEGO sets, each with attributes such as release year, theme, number of pieces, original retail price (USD_MSRP), and current resale price (Current_Price). To ensure data quality and improve model accuracy, we applied filtering techniques to remove data points with missing values for ‘USD_MSRP,’ resulting in a cleaned dataset of 3,607 complete entries. This refined dataset allows for a robust regression analysis, as it captures key factors influencing LEGO resale prices while minimizing data inconsistencies. Additionally, by leveraging data from trusted sources like Brick-

set and Bricklink, which are widely recognized in the LEGO collector community, we ensure the validity and reliability of our study. The structured nature of the dataset makes it well-suited for machine learning applications, allowing us to build an accurate, data-driven model for predicting LEGO resale prices.

Our project focuses on developing a predictive model for estimating the resale price of LEGO sets based on their core attributes. Since the release of our proposal and milestone papers, we have significantly expanded our modeling efforts while maintaining the same ultimate goal: to provide a reliable, data-driven tool that collectors, investors, and resellers can use to forecast resale values more precisely than subjective appraisals or historical lookups. One major challenge in building this model is the inherent complexity of the LEGO secondary market. Resale prices are influenced not only by structured attributes like set size, release year, and MSRP but also by unstructured external factors such as media tie-ins, nostalgia, and production scarcity. For example, sets tied to major franchises like Star Wars or Harry Potter often exhibit appreciation trends that are difficult to capture with simple linear models. These complexities highlight the need for more sophisticated modeling techniques.

To address these challenges, we filtered and preprocessed the dataset, reducing it from 14,936 entries to 3,607 complete records. We retained the most critical attributes: ‘Year,’ ‘Theme,’ ‘Pieces,’ ‘USD_MSRP,’ and ‘Current_Price.’ Through exploratory data analysis (EDA), we identified that ‘Pieces’ and ‘USD_MSRP’ were the strongest predictors of resale value, while ‘Year’ exhibited a weaker negative correlation. Although ‘Theme’ is a categorical variable, it proved to be highly influential when encoded appropriately, particularly for popular franchises.

We implemented and evaluated several machine learning models to predict resale prices. Our first model was a Linear Regression model, which we used to establish a baseline for evaluating more complex techniques. We then trained Decision Tree Regressors (with max depth tuned to 7), Random For-

est Regressors (with max depth tuned to 11), and Gradient Boosting Regressors (optimized with approximately 30 boosting iterations). Each model was evaluated using 5-fold cross-validation, where we recorded Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 across folds to assess both performance and consistency.

To isolate the impact of categorical variables, we also experimented with a reduced feature set that excluded ‘Theme’ and its associated one-hot encodings. This led to a significant drop in R^2 , confirming that theme-related information plays a substantial role in predicting LEGO resale prices. We benchmarked all machine learning models against trivial baselines, including models that simply predicted the mean resale price, the median resale price, or generated random predictions sampled from a Gaussian distribution matching the dataset’s mean and standard deviation. Our best-performing models – particularly the Decision Tree and Gradient Boosting regressors – substantially outperformed all baselines across every evaluation metric.

Although our initial project plan included exploring Support Vector Regressors (SVRs) and Multilayer Perceptrons (MLPs), we deprioritized these methods after preliminary SVR tests showed poor performance and due to limited time for deep model development. Additionally, while we identified external data sources such as eBay sales histories and discussed potential natural language processing (NLP) applications for analyzing LEGO set names (e.g., detecting “Collector’s Edition” sets), we deferred these ideas for future research. Overall, our current work prioritizes ensemble methods that demonstrate a greater capacity for capturing the nonlinear dynamics of the LEGO resale market, setting a strong foundation for further refinements in future iterations of this project.

Based on our experiments, we observed that simple linear models could capture only limited trends in LEGO resale prices, yielding relatively high error metrics and moderate R^2 scores. More sophisticated, nonlinear models – particularly Decision Trees and Gradient Boosting – substantially improved predic-

tive accuracy, demonstrating R^2 values near or above 0.8. Feature importance analysis showed that ‘Pieces’ and ‘USD_MSRP’ were strong drivers of resale value, while theme-level distinctions captured important variance related to branding and cultural significance. Removing ‘Theme’ significantly worsened model performance, confirming its critical role. Additionally, the superiority of tree-based models over simple baselines and linear regression indicates that LEGO resale value prediction involves complex, nonlinear interactions between features. Moving forward, our work suggests that ensemble techniques and careful feature engineering are crucial for modeling collectible markets like LEGO. In future research, we aim to incorporate external data sources, explore more advanced ensemble methods, and potentially apply deep learning approaches if sufficient data augmentation is possible.

2 Related Work

The growing interest in LEGO as an alternative investment asset has motivated several financial and economic analyses exploring its potential return on investment. Dobrynska and Kishilova (2022) present one of the most comprehensive quantitative studies in this space, analyzing 2,322 discontinued LEGO sets from 1987 to 2015. They report that LEGO sets yield impressive average annual returns of 11%, surpassing traditional assets like stocks, bonds, and gold – a point also noted in Section 1. Importantly, they construct LEGO-specific price indices and identify that rare, small, and licensed sets (especially Star Wars and Architecture) generate the highest appreciation. Similarly, Fan (2022) studies 838 LEGO sets and draws parallels between LEGO and other collectible markets like art and wine, highlighting the role of scarcity, age, and fan engagement in price growth. Both studies emphasize a financial framework – using APR, HPR, and hedonic modeling – to assess investment viability. While these works provide compelling evidence that LEGO sets behave like appreciating assets, they stop short of offering tools to predict future prices based on set features. Our project fills this gap by building a pre-

dictive machine learning model trained on structured LEGO features, enabling forward-looking price estimation and giving collectors and investors a tool to assess resale value before purchasing, rather than analyzing ROI after the fact. This focus on prospective decision support, rather than retrospective evaluation, makes our work more immediately actionable and investment-driven.

In contrast to purely financial analysis, Tangeric’s capstone project (n.d.) directly explores machine learning approaches for LEGO price prediction. Using the same Kaggle dataset that our team uses – “LEGO Sets and Prices Over Time” – Tangeric expands it with additional data from Bricklink and Brickset APIs and applies multiple models including Lasso, Random Forest, XGBoost, and neural networks to predict current resale prices. XGBoost achieves the best results (MAE under \$15), outperforming linear methods and deep learning models. This project demonstrates the viability of ensemble methods and highlights the importance of preprocessing, one-hot encoding of themes, and careful outlier handling. Inspired by Tangeric’s approach, our project similarly evaluates multiple regression models, including Random Forest and decision trees, but improves upon his work by incorporating correlation analysis, proximity measures (e.g., Minkowski distance), and expanded feature engineering to enhance model interpretability and performance. Unlike Tangeric’s model, which focuses mainly on predictive accuracy, our project emphasizes both performance and explainability – using feature correlation matrices, distance metrics, and exploratory data visualizations to interpret what factors drive price appreciation. This dual emphasis not only supports model development but also empowers end users to make informed investment decisions grounded in transparent, interpretable insights.

A complementary perspective is provided by Codignola et al. (2021), who analyze the emergence of Adult Fans of LEGO (AFOLs) as a core consumer demographic. Their work demonstrates how LEGO’s shift toward adult-oriented themes and complex builds has cultivated a strong aftermarket fueled by nostalgia, identity, and community engagement.

The study outlines how adult collectors drive the demand behind secondary market appreciation and influence set design and marketing strategies. While their findings are qualitative and sociological in nature, they illuminate why certain themes (e.g., Modular Buildings, Star Wars) may have a higher resale trajectory than others. Our project builds on this insight by operationalizing it – embedding sociocultural variables like theme, year, and set complexity into quantitative models. By integrating the behavioral drivers that Codignola et al. identify into structured data representations, we create a modeling framework that blends economic behavior with predictive analytics, making our work uniquely interdisciplinary and empirically grounded.

Finally, Peterson and Ziegler (2021) approach LEGO resale data from an educational perspective, designing a classroom activity to teach multiple linear regression. Their curriculum involves modeling resale price with predictors like set size and theme, helping students learn statistical concepts like dummy variables and interaction effects. While their scope is limited to three themes and the pedagogical goal is conceptual understanding, their work affirms that LEGO data is a rich resource for statistical learning. Our work takes the foundational modeling approach outlined in their project and extends it in scale, complexity, and purpose – transitioning from classroom demonstrations to a full-scale, k-fold validated, real-world predictive model. Where Peterson and Ziegler emphasize accessibility and simplicity, our focus is on robustness, generalizability, and application to the investment decision-making process. This evolution from education to deployment demonstrates the practical potential of LEGO data beyond academic exercises.

Together, these studies offer a rich foundation of financial, sociocultural, and technical insights on LEGO’s resale ecosystem. What distinguishes our work is its emphasis on interpretable machine learning models designed to support real-world pricing decisions – not just predicting values, but surfacing the why behind them. Through enhanced data cleaning, exploratory analysis, and model evaluation, our project aspires to deliver a replica-

ble, scalable, and user-informative framework that equips LEGO collectors and investors with predictive tools tailored to a dynamic and growing secondary market.

3 Problem Definition

3.1 Datasets Description

The dataset used in this study is titled “LEGO Sets and Prices Over Time”, and it was sourced from Kaggle. It was compiled by Alex Racape using a combination of data from two public APIs: Brickset.com and Bricklink.com. The Brickset API provided list prices and metadata for LEGO sets, while the Bricklink API contributed the most recent resale prices. Both sources were queried on May 10, 2023, making the dataset relatively current. Because the dataset was created and shared by an external contributor, it is not an original dataset developed by our team. The full dataset is available for public download at: <https://www.kaggle.com/datasets/alexracape/lego-sets-and-prices-over-time/data>.

The dataset is stored in tabular format as a CSV file named ‘sets.csv.’ Each row corresponds to a unique LEGO set, and each column represents a different feature or attribute of the set. In total, the dataset includes 14,936 distinct data points, representing individual LEGO products. The original dataset includes 17 attributes: ‘Set ID,’ ‘Name,’ ‘Year,’ ‘Theme,’ ‘Theme Group,’ ‘Subtheme,’ ‘Category,’ ‘Packaging,’ ‘Number of Instructions,’ ‘Availability,’ ‘Pieces,’ ‘Minifigures,’ ‘Owned,’ ‘Rating,’ ‘USD_MSRP,’ ‘Total Quantity,’ and ‘Current Price.’ These attributes span a mix of categorical (e.g., ‘Theme,’ ‘Category’), numerical (e.g., ‘Pieces,’ ‘USD_MSRP’), and ordinal data types (e.g., ‘Year’).

To prepare the dataset for use in our regression model, we filtered out entries with missing values in critical fields such as ‘USD_MSRP.’ This step removed a large portion of the dataset but left us with a high-quality subset of 3,607 complete data points. From there, we retained only the most relevant features: ‘Name,’ ‘Year,’ ‘Theme,’ ‘Pieces,’ ‘USD_MSRP,’ and ‘Current Price.’ These attributes were selected based on their expected

impact on resale value and their completeness in the dataset. As a result, the cleaned and filtered dataset is well-suited for modeling LEGO resale prices based on structured product metadata, summarized in Table 1 below:

Attribute	Type	Most Popular Value	Meaning
Name	Categorical (Object)	”Star Wars Advent Calendar”	Name of LEGO Set
Year	Numerical (Integer)	2019	Year LEGO Set Was Released
Theme	Categorical (Object)	”Star Wars”	Broad Category Set Belongs To
Pieces	Numerical (Float)	1.0	Number of Pieces in Set
USD_MSRP	Numerical (Float)	9.99	Manufacturer’s Suggested Retail Price, USD
Current_Price	Numerical (Float)	20.0	Most Recent Resale Price, USD (Target Variable)

Table 1: *Summary of Attributes in Filtered Dataset*

For our regression model, we rely on a single dataset – the filtered version titled ‘reduced_lego_data.csv’ – which contains the key attributes required to predict ‘Current Price’ without any missing values for ‘USD_MSRP.’ The original dataset (sets.csv) included many incomplete entries, particularly missing values in the ‘USD_MSRP’ field. During pre-processing, we removed these entries to ensure data integrity and modeling accuracy. The resulting filtered dataset is clean and well-structured, containing the most relevant features such as ‘Name,’ ‘Year,’ ‘Theme,’ ‘Pieces,’ ‘USD_MSRP,’ and ‘Current Price.’ Because this dataset is already complete and self-contained, there was no need to integrate multiple datasets or align data points and attributes. We have also downloaded the raw data files for analysis and backup. The origi-

nal CSV file (sets.csv) is approximately 1,741 KB (1.78 MB) in size, and when compressed into a ZIP file for data sharing, the size is reduced to 368 KB.

3.2 Task Definition

After filtering and preparing the dataset, we have chosen ‘Current_Price’ as the prediction target (label). This attribute represents the most recent resale price of a LEGO set in USD. Our goal is to build a regression model that accurately predicts ‘Current_Price’ based on features such as ‘Year,’ ‘Theme,’ ‘Pieces,’ and ‘USD_MSRP.’ This will allow us to estimate the expected market value of LEGO sets based on historical trends and set characteristics. We are focusing on a single prediction task – estimating ‘Current_Price.’ While multi-task learning could involve predicting additional attributes (e.g., estimating popularity or rating alongside price), our primary objective is to develop a robust regression model for price prediction.

The label ‘Current_Price’ is a continuous numerical variable, representing a dollar amount. Because it is not categorical, our prediction task is regression rather than classification. However, if we were to transform ‘Current_Price’ into discrete price ranges (e.g., low, medium, high resale value), the problem would shift to a multi-class classification task. Since our goal is to predict exact resale values, regression is the most appropriate approach. For our regression model, the most relevant features for predicting ‘Current_Price’ are ‘Year,’ ‘Theme,’ ‘Pieces,’ and ‘USD_MSRP,’ as they directly influence a LEGO set’s resale value. While ‘Name’ may contain useful information, it requires additional processing to extract meaningful insights. The table below summarizes each attribute’s potential contribution to our model:

After filtering out null values and missing data, we have 3,607 complete data points available for modeling. This cleaned dataset ensures that our regression model is trained on high-quality, complete data without gaps or inconsistencies.

Name	Feature?	Reasoning
Name	Possibly, but Limited	While set name may contain useful information (e.g., keywords) it is mostly unstructured text. Natural language processing (NLP) techniques required to extract insights.
Year	Yes	Older LEGO sets tend to appreciate in value.
Theme	Yes	Some themes (e.g., Star Wars, Harry Potter) hold higher value in the resale market.
Pieces	Yes	Larger sets often have higher resale prices, making the number of pieces a strong numerical predictor.
USD_MSRP	Yes	The original retail price is a key indicator of resale value, as higher MSRP sets generally resell for more.

Table 2: *Feature Selection for LEGO Resale Price Prediction*

4 Data Analysis

4.1 Analyzing Individual Attributes (Including Features & Label)

As discussed in the previous section, the most relevant attributes for predicting ‘Current_Price’ are ‘Year,’ ‘Theme,’ ‘Pieces,’ and ‘USD_MSRP.’ Among these, ‘Theme’ is a categorical attribute that represents the collection under which a LEGO set was originally marketed. Some of the most popular LEGO themes include Star Wars, City, and Friends, which have not only inspired bestselling sets but have also expanded into television shows, movies, and video games, contributing to their cultural significance. To analyze the prevalence of different themes in our dataset, we created a bar plot illustrating their respective popularity. Since there are 97 unique themes, we focused on those with at least 45 sets to ensure meaningful observations. As shown in Figure 1, a few themes dominate the dataset with hundreds of sets, while many others have only a handful of entries. This distribution suggests that themes with a larger number of sets may contribute more reliable patterns for prediction, though further analysis is needed to confirm this.

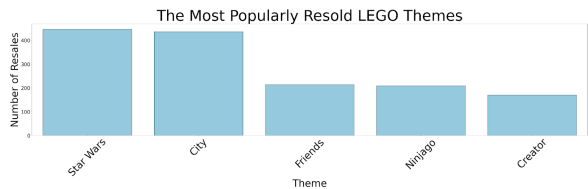


Figure 1: *Bar Plot of the Most Popular LEGO Themes*

To further analyze the attributes in our dataset, we generated histograms for each numerical feature. One of the key attributes we examined was ‘Pieces,’ which represents the number of pieces included in a LEGO

set. The resulting histogram revealed a right-skewed distribution, indicating that while most LEGO sets contain a relatively small number of pieces, a few sets have significantly larger piece counts. This suggests that while smaller sets are more common, larger sets – often collector’s editions or complex builds – are outliers in terms of piece count. Figure 2 illustrates this distribution below.

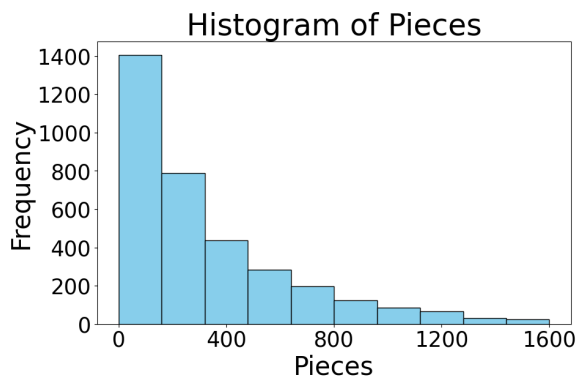


Figure 2: *Histogram of LEGO Set Piece Counts*

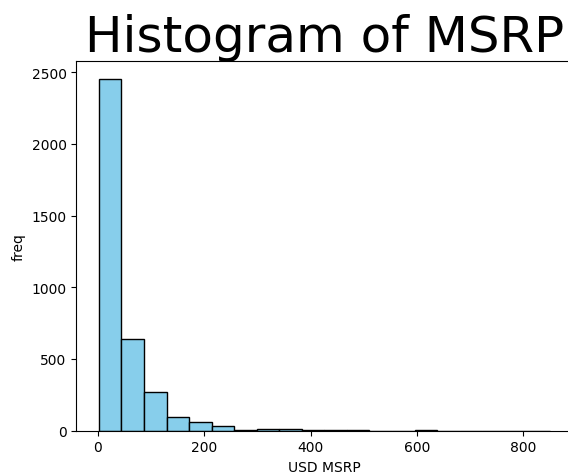


Figure 3: *Histogram of MSRP*

LEGO sets are known for their relatively high prices within the toy industry, yet the majority have modest market values. Figure 3 above illustrates the distribution of original retail prices (MSRP), showing that most LEGO sets were initially sold at affordable price points. Similarly, the resale market for LEGO appears to be thriving, but as shown in Figure 4, most transactions occur at moderate prices rather than at extreme highs. As expected, the distributions of MSRP and resale prices share a similar pattern, suggesting a correlation between the two. However, external fac-

Histogram of Current Resale Price

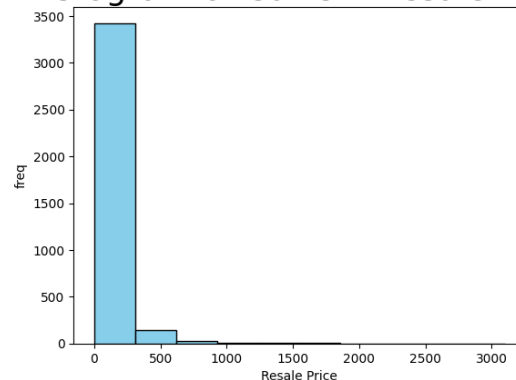


Figure 4: *Histogram of Resale Prices*

tors such as scarcity, cultural trends, and nostalgia can significantly drive up resale prices for certain sets. This introduces an intriguing challenge for machine learning, as predicting resale value involves uncovering the hidden influences that cause some sets to appreciate more than others.

Finally, the release year of each LEGO set may influence its resale price. Over time, LEGO has grown in popularity, and the data reflects an increasing number of sets being produced in recent years. However, the impact of this trend on the resale market remains uncertain. While newer sets are more abundant, older sets may hold greater resale value due to their rarity and obsolescence. Figure 5 presents the distribution of release years, highlighting the shift in production volume over time.

Histogram of Release Year

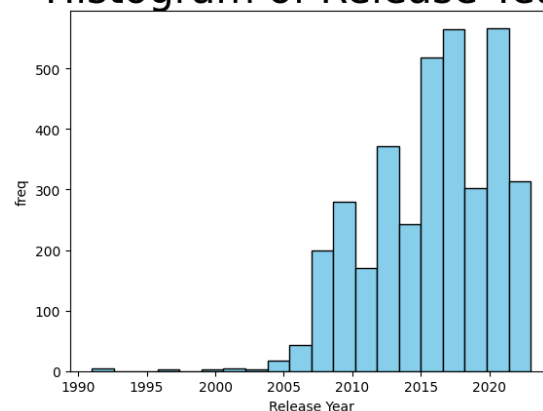


Figure 5: *Histogram of Release Year*

Since much of our dataset consists of numerical attributes, we examined their descriptive

statistics to uncover key insights. Some extreme values stood out, such as a maximum of 11,695 pieces, an MSRP of \$849.99, and a resale price reaching \$3,099.99. These figures suggest the presence of outliers, which are evident in the corresponding histograms. While these outliers may skew the overall distribution, they provide valuable insights into consumer behavior, particularly regarding high-end collector's sets. Table 3 below presents the full descriptive statistics for our dataset.

	Year	Pieces	USD_MSRP	Current_Price
Count	3607	3607	3607	3607
Mean	2015.704186	444.148877	46.279795	86.038420
STD	4.590598	714.511411	61.502246	146.239004
Min	1991	0	1.99	0.05
25%	2013	92	12.99	19.99
50%	2016	233	29.99	40
75%	2019	514	49.99	95
Max	2023	11695	849.99	3099.99

Table 3: *Descriptive Statistics of Key LEGO Set Attributes*

4.2 Analyzing Proximity Between Data Points

Analyzing the cosine similarity of our dataset revealed some unexpected patterns. Due to the large size of the dataset, we conducted a focused analysis on a small sample of 10 data points, generating a two-dimensional matrix that measures the similarity between each pair of points. Surprisingly, many data points that we initially assumed to be unrelated exhibited high similarity scores. While the implications of this finding are not yet fully clear, it is an important observation to consider before training our models. Below in Table 4, we present the cosine similarity matrix for the selected data points, where each number along the top and left axes represents a feature in the dataset. A cosine similarity score of 1.0000 indicates that a feature is being compared to itself.

	13449	12053	11757	13648	10905	7012	9762	14589	10232	7500
13449	1	.268713	.456222	.385117	.534499	.251778	.396914	.572753	.269782	.352327
12053	.268713	1	.979665	.992407	.956992	.999839	.987493	.943399	.999972	.995709
11757	.456222	.979665	1	.996837	.995605	.976026	.995605	.990687	.979935	.993393
13648	.385117	.992407	.996837	1	.985050	.990080	.996589	.976994	.992497	.998897
10905	.534499	.956992	.995605	.985050	1	.951901	.987088	.998460	.957535	.979220
7012	.251778	.999839	.976026	.990080	.951901	1	.985220	.937428	.990819	.994107
9762	.396914	.987493	.995605	.996589	.987088	.985220	1	.977138	.988194	.997530
14589	.572753	.943399	.990687	.976994	.998460	.937428	.977138	1	.943757	.968615
10232	.269782	.999972	.979935	.992497	.957535	.990819	.988134	.943757	1	.995989
7500	.352327	.995709	.993393	.998897	.979220	.994107	.997530	.968615	.995989	1

Table 4: *Cosine Similarity Matrix For Sampled LEGO Sets*

Next, we examined the Minkowski distances

between all data points to measure their relative dissimilarity. The Minkowski distance is a flexible metric that generalizes various distance measures, including Manhattan ($p = 1$) and Euclidean ($p = 2$). Additionally, extremely high Minkowski distances can help identify potential outliers within the dataset. To gain a global perspective on how data points are distributed, we computed the average Minkowski distance for each point relative to all others. We then visualized these distributions using histograms for three different values of p : $p = 1$ (Manhattan Distance), $p = 2$ (Euclidean Distance), and $p = 50$ (approximating the Chebyshev Distance). Figures 6, 7, and 8 illustrate these distributions, providing insight into how data points vary in proximity and highlighting any potential anomalies.

Histogram of Minkowski Distances ($p=1$)

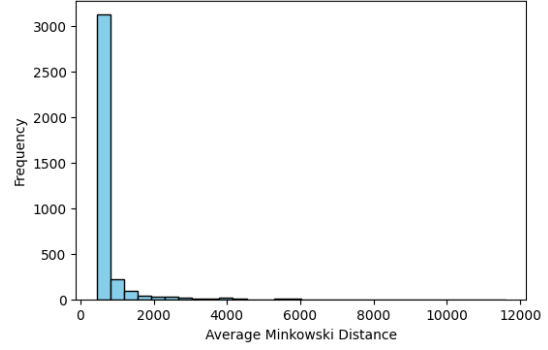


Figure 6: *Distribution of Manhattan Distances ($p=1$) Across Data Points*

Histogram of Minkowski Distances ($p=2$)

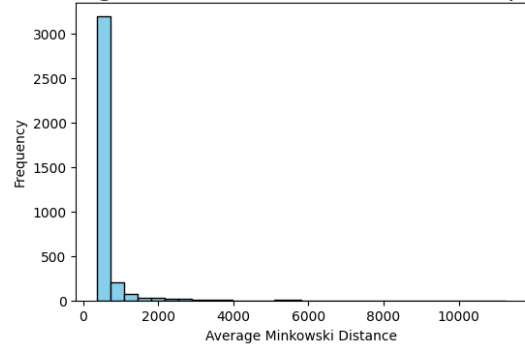


Figure 7: *Distribution of Euclidean Distances ($p=2$) Across Data Points*

Each of these histograms exhibits a right-skewed distribution, indicating that most data points are clustered closely together, while a few outliers are significantly distant from the rest.

Histogram of Minkowski Distances ($p=50$)

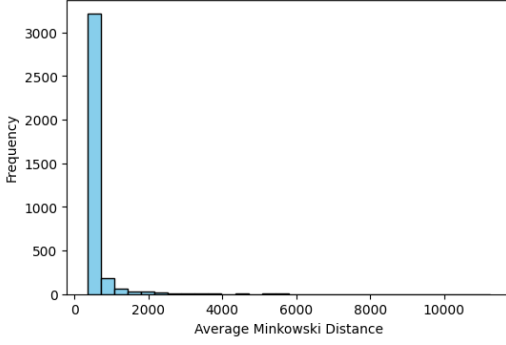


Figure 8: *Distribution of Chebyshev Distances ($p=50$) Across Data Points*

4.3 Analyzing Correlations (Feature vs Label)

To examine the relationships between our numerical features, we calculated the Pearson Correlation Coefficient to quantify their correlations. The matrix in Table 5 below provides insights into the strength and direction of these relationships.

	Current_Price	USD_MSRP	Pieces	Year
Current_Price	1	.618180	.567148	-.220335
USD_MSRP	.618180	1	.893495	.113496
Pieces	.567148	.893495	1	.121926
Year	-.220335	.113496	.121926	1

Table 5: *Pearson Correlation Matrix For Numerical Features*

Each value of 1.000 in the matrix represents the correlation of a feature with itself. Beyond these expected values, the coefficients reveal several interesting insights. Notably, there is a strong positive correlation of approximately 0.89 between Pieces and USD_MSRP, which aligns with expectations from a business standpoint – larger LEGO sets generally have higher retail prices. Additionally, there is a weak but notable negative correlation between Year and Current Resale Price, indicating that older sets tend to sell for slightly higher prices. This trend suggests that nostalgia may play a role in influencing the purchasing decisions of collectors and resellers in the secondary LEGO market.

4.4 Encoding Categorical Features, If Necessary

Our filtered dataset contains only one categorical attribute: Theme. We hypothesize

that Theme plays a significant role in determining a set’s Current Resale Price. Therefore, encoding this feature is crucial to understanding its impact and relationship with price fluctuations. Since LEGO Themes have no inherent ordinal structure, we will use **One-Hot Encoding** to represent them. Despite the large number of unique themes, we do not expect this encoding method to have a significant negative effect on model performance, as it allows us to capture meaningful distinctions between different LEGO themes without imposing an arbitrary ranking.

5 Validation and Evaluation Methods

5.1 Validation Method (Experimental Design)

For our project, we will use a train-test split instead of a train-validation-test split. We will choose a train-test split over a train-validation-test split because our dataset, after filtering, consists of only 3,607 data points. Introducing a separate validation set would further reduce the training data available for model learning, potentially weakening generalization.

To maximize the efficiency of our train-test split for each model, we used hold-out validation. Hold-out validation refers to splitting the available dataset into two distinct subsets: one for training the model and one for testing its performance. This strategy allowed us to iteratively test each model with five different train-test split ratios: 50-50%, 60-40%, 70-30%, 80-20%, and 90-10% for the training and testing, respectively. For each machine learning model we tested on the data, we selected the most successful split before refining that model’s hyperparameters. As such, hold-out validation allowed for more flexibility than cross-validation, because we wanted to select specific and incremental training percentages.

5.2 Evaluation Metrics

Since our task is regression-based, we will not use classification metrics such as accuracy, precision, recall, or F1-score. Instead, we will evaluate model performance using standard regression metrics that measure the accuracy and reliability of our predictions. The

first metric we will use is Mean Absolute Error (MAE), which calculates the average absolute difference between predicted and actual resale prices. A lower MAE indicates that the model’s predictions are closer to actual values, making it a key metric for measuring overall prediction accuracy. Additionally, we will use Root Mean Squared Error (RMSE), which penalizes larger errors more heavily than MAE. RMSE is particularly useful for detecting significant mispredictions, as it emphasizes large deviations from actual prices. Like MAE, a smaller RMSE is preferred, as it indicates fewer large prediction errors.

To measure how well our model explains the variability in LEGO resale prices, we will also use R-Squared (R^2 Score), which evaluates the proportion of variance in resale price that our model accounts for, with values closer to 1 indicating a strong explanatory model. If R^2 is close to 0, it suggests that our model does not effectively capture the relationship between features and price.

By using these evaluation metrics, we will assess both prediction accuracy (MAE, RMSE) and model explanatory power (R^2) to compare different models. The model that produces the most accurate and generalizable predictions will be selected as the final approach for estimating LEGO resale prices.

5.3 Baselines

To evaluate the effectiveness of our model, we will compare its performance against several baseline approaches. Since our task involves predicting the resale price of LEGO sets, we will use simple, non-machine-learning baselines as benchmarks to ensure that our model provides meaningful improvements. The first baseline will be the mean price baseline, which predicts every LEGO set’s resale price as the average resale price found in the training data. While simplistic, this serves as a foundational benchmark that our model should outperform. Similarly, we will use a median price baseline, which predicts all resale values as the dataset’s median price. This approach helps mitigate the influence of extreme outliers and provides a more robust point of comparison.

In addition to statistical baselines, we will introduce a random guessing baseline, where resale prices are generated by randomly sampling from a Gaussian distribution with the same mean and standard deviation as the training data. Although not a realistic model, this will serve as a lower-bound benchmark. Lastly, we will use a linear regression baseline, which assumes a direct linear relationship between features such as the year of release, theme, number of pieces, and MSRP, and the resale price. This will be our first machine learning benchmark before we explore more complex models like decision trees, random forests, or gradient boosting techniques.

Our evaluation will focus on MAE, RMSE, and R^2 metrics, ensuring that our final model provides more accurate and reliable predictions than these baselines. If our model does not significantly outperform these simple methods, we will refine our train-test split, hyperparameter tuning, or model complexity to improve performance. By comparing against well-defined baseline models, we will validate the effectiveness of our approach in predicting LEGO resale prices with higher accuracy and generalizability.

5.4 Comparison Design

We put a lot of thought and effort into selecting the best configuration of each machine learning model we used on the data to predict resale LEGO prices. As stated in section 5.1 on validation methods, we determined the unique train-test split for each model that made it perform the best. For the three best models, we also performed hyperparameter tuning. For these models, we graphed the RMSE of the model against a value of a particular hyperparameter of that model to visually and mathematically observe how we should configure a given model. For models such as Gradient Boosting, we decided to tune for the best number of iterations to run the algorithm. Meanwhile, for our Random Forest and Decision Tree Regressors, we tuned the model for the best value for maximum depth. Once we felt each model achieved its own optimal performance, we compared each model based on the evaluation and regressions metrics outlined in section 5.2: MAE, R^2 , and RMSE.

We graphed these results with matplotlib and organized the results neatly into a table, as seen in section 7.

6 Data-Driven Methods

To build a model capable of accurately predicting LEGO set resale prices, we began by preparing and transforming our dataset. We identified the most important features by examining both the raw dataset and conducting a correlation analysis. The attributes we used include ‘Year,’ ‘Pieces,’ ‘USD_MSRP,’ and ‘Theme.’ ‘Year’ captures the release date of the set, which may reflect rarity or nostalgia-driven value. ‘Pieces’ is a direct proxy for set size and complexity. ‘USD_MSRP’ (Manufacturer’s Suggested Retail Price) represents the original retail value, which we hypothesized to be strongly correlated with resale value. ‘Theme’ is a categorical attribute indicating the franchise or category of the set, such as Star Wars or City, which we expected to influence collector demand. We also engineered two additional features: ‘Price_per_piece’ (calculated as MSRP divided by number of pieces) and ‘Price_appreciation’ (resale price divided by MSRP). While useful for understanding price dynamics, these engineered features were excluded from model training to prevent data leakage, since they are both derived from the label variable (Current_Price).

To encode the ‘Theme’ variable, we applied one-hot encoding, transforming the categorical column into multiple binary columns representing the presence or absence of each theme. This allowed us to include theme data in models that require numerical input, such as linear regression. Although this expanded the feature space significantly due to the large number of unique themes, we retained all binary columns in our initial experiments. While we originally considered using dimensionality reduction techniques such as Principal Component Analysis (PCA) to compress the high-dimensional theme space, we ultimately decided not to implement PCA after early experiments showed the full feature set was manageable and interpretable.

We performed a correlation analysis on the numerical features to assess how each one re-

lates to the target variable, ‘Current_Price.’ Pearson correlation coefficients revealed that ‘USD_MSRP’ and ‘Pieces’ are positively correlated with resale price, while ‘Year’ is weakly negatively correlated. This insight aligns with economic intuition: larger and more expensive sets tend to retain or appreciate value, while older sets may become rarer and more desirable to collectors. Although we began by using the full feature set, we later compared results with a reduced feature configuration (excluding ‘Theme’) to better understand its contribution to model accuracy. The reduced model showed a drop in R^2 , suggesting that theme-based branding plays a significant role in predicting resale value.

With our features finalized, we evaluated five machine learning models: Linear Regression, Decision Tree Regressor, Random Forest Regressor, Gradient Boosting Regressor, and Support Vector Regressor (SVR). Initially, we used 5-fold cross-validation to measure generalizability. However, after consulting with Professor Meng Jiang, we transitioned to a hold-out validation approach to refine our train-test split ratio. This allowed us to better control the randomization of training and test splits and observe variation in performance across different splits. Each model was evaluated using three regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2). These metrics allowed us to assess both prediction accuracy and explanatory power. We used the hold-out validation results to guide hyperparameter tuning for each model. Specifically, for the Decision Tree, we tuned the maximum depth (best around 7); for the Random Forest, we also tuned the maximum depth (best around 11); and for the Gradient Boosting Regressor, we found optimal performance with approximately 30 iterations. We deliberately did not tune the Linear Regression model, choosing instead to keep it simple and interpretable as our first model. Similarly, we did not tune the SVR due to its extremely poor performance on RMSE during initial trials, which made further experimentation with that model unnecessary.

To contextualize our machine learning models, we implemented three trivial baselines: a con-

stant predictor using the mean resale price, a median price predictor, and a Gaussian random predictor using the same mean and standard deviation as the dataset. These baselines served as controls to ensure that our models were learning meaningful patterns rather than mimicking simple heuristics. Across all metrics, our best models – particularly Decision Tree and Gradient Boosting – substantially outperformed the baselines. Finally, we visualized how well our predictions aligned with true values by plotting predicted vs. actual resale prices. We also compared them to the constant mean baseline to assess deviation across the price range. These plots highlighted how tree-based models better captured resale variation for both lower- and higher-value sets. While we have not yet implemented model interpretability techniques such as SHAP values, we recognize their value and plan to incorporate them into future work.

In summary, our data-driven modeling process began with correlation-based feature evaluation, one-hot encoding for categorical data, and structured experimentation with five regression algorithms. After testing both full and reduced feature sets, and switching from cross-validation to hold-out validation for better experimental control, we concluded that ensemble tree methods performed best. In future phases, we plan to investigate additional model tuning, interpretability tools, and potentially external feature enhancements to further improve accuracy and usability.

7 Experimental Results

In our experiment, we tested a total of five models and three baselines, and analyzed their performance. The first model we tested in-depth was a simple Linear Regression model. The results from our evaluation reveal several key insights, most notably that the dataset does not exhibit strong linear relationships between the features and the target variable. The Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) are both relatively high, especially when considered in the context of the dataset’s average resale price of approximately \$86. These large error values point to suboptimal performance from the default linear regression model.

Given these indications of non-linearity, we also sought to better understand the model’s behavior by visualizing its predictions on a small subset of the data. Although not exhaustive, this type of qualitative analysis offers a clearer picture of where the model succeeds and where it struggles. As shown in Figure 9, we plotted seven sample points, comparing the predicted resale prices (in red) with their actual values (in blue).



Figure 9: *Predicted vs Actual Resale Prices for Sample LEGO Sets*

As shown in the plot, the linear regression model performs reasonably well when predicting resale prices in the range of \$50 to \$100. In this mid-range, the predicted values closely align with the actual prices. However, as the actual resale price increases, the model’s predictive accuracy begins to decline significantly. The most pronounced example of this can be seen with the LEGO set labeled “Titanic,” which exhibits the largest residual between its predicted and actual values – highlighting the model’s inability to capture higher-end price dynamics.

In response to the model’s overall underperformance, we hypothesized that a simpler linear model – trained on a reduced feature set – might yield comparable results. To identify low-impact features, we conducted a correlation analysis, which revealed that the variable “Theme” generally had low correlation with the target variable, especially for themes with few data points in the dataset. Based on this insight, we removed the “Theme” attribute along with its associated one-hot encoded features. This reduction simplified the dataset and potentially reduced overfitting.

We then retrained the default linear regression model using this feature-reduced dataset and re-evaluated it using the same metrics – MAE, RMSE, and R-squared – to assess performance. The results of this evaluation are presented in Table 6.

Reduced Linear Model	Value
Mean Absolute Error	47.135
Root Mean Squared Error	87.768
R-Squared	0.299

Table 6: *Feature-Reduced Linear Regression Model (Theme Removed)*

The evaluation metrics from the reduced linear model – particularly the MAE and RMSE – are comparable to those from the original model. However, the most notable difference lies in the R-squared value, which dropped from 0.59 to 0.299. This decline suggests that removing the “Theme” feature significantly impacted the model’s ability to explain variance in resale price. Although “Theme” showed low correlation with the target variable overall, its aggregate predictive value appears to be meaningful, especially when considering the full diversity of LEGO sets. This indicates that even features with many unique values and sparse representation can contribute valuable context to a regression model. Given that including “Theme” does not impose a heavy computational burden, it should remain part of the feature set in future model iterations.

Across all evaluation metrics, the linear regression model consistently outperforms the trivial baseline models, reinforcing that it captures meaningful relationships within the data rather than relying on simple statistical heuristics. Nonetheless, the model’s relatively high error values and moderate R-squared indicate that it struggles to fully capture the complexity of the LEGO resale market. To address this, we extended our analysis by experimenting with additional regression models. We tested Decision Trees, Random Forests, Support Vector Regression (SVR), and Gradient Boosting to better model potential non-linear relationships. Through these methods, we hoped to enhance predictive accuracy and

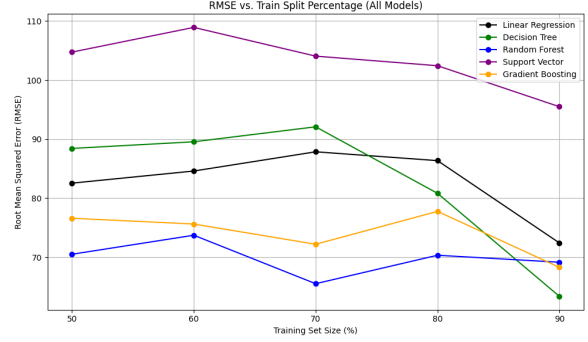


Figure 10: *RMSE vs. Train Split Percentage for Linear and Nonlinear Models*

more effectively model the nuanced factors that influence LEGO set resale prices. As explained in Section 5.4 on comparison design, we first sought to optimize the train-test split for each of our new models. In the graph in Figure 10, we plotted the performance in RMSE for each model across five different training set splits (50% to 90%), which provided a broader view of generalizability and supported hyperparameter tuning. For most splits, the Random Forest Regressor had the best results, but for particular splits like the 90-10%, both the Decision Tree and Gradient Boosting algorithms outperformed the Random Forest.

This graph was crucial to our completion of the remainder of the project for a few reasons. Firstly, it showed us what models were successful and seemed like good candidates for hyperparameter tuning. Because we simply used the default version of each model to complete this graph, there was certainly better performance that could result from our hyperparameter analysis. Secondly, it also taught us which models did not work well on our data. As can be seen by the line plot labeled for the SVR, this algorithm performed the poorest on all train-test splits. As such, we decided to abandon the tuning of this model in favor of focusing on the more promising ones: Decision Tree, Random Forest, and Gradient Boosting.

Thus, we performed hyperparameter tuning to optimize these most promising models. For the Decision Tree, we tested max_depth values ranging from 2 to 20 and found the lowest RMSE at a depth of 7, as shown in Figure 11. Similarly, Random Forests performed best at

a depth of 11 (Figure 12), and Gradient Boosting achieved its best results with 30 iterations (Figure 13). Beyond these thresholds, performance plateaued or worsened due to overfitting.

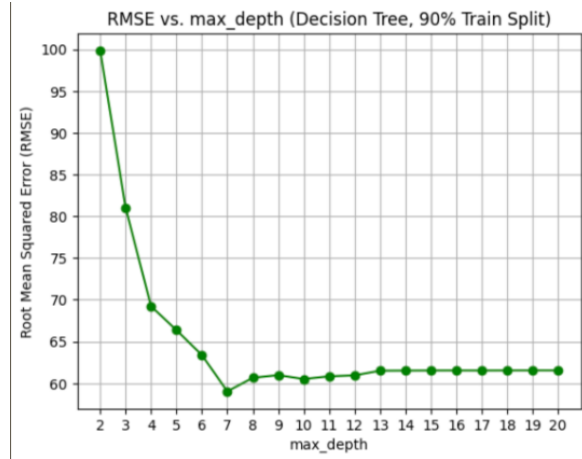


Figure 11: *Decision Tree RMSE vs. Max Depth*

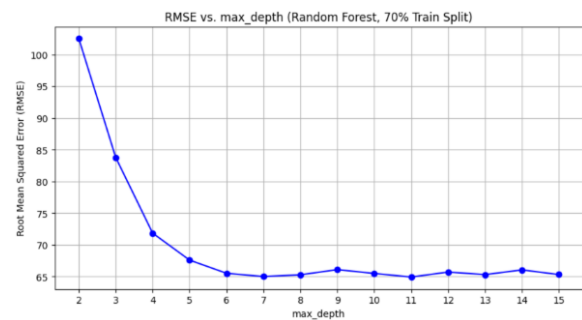


Figure 12: *Random Forest RMSE vs. Max Depth*

We evaluated each model using three standard regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2). Table 8 reports the best performance observed for each model across its optimal train/test split. The Decision Tree Regressor (depth = 7) achieved the lowest RMSE (59.0294) and highest R^2 (0.8011), outperforming all other models and baselines. While the Random Forest and Gradient Boosting models also performed well, the simpler Decision Tree provided the best balance of performance and interpretability. Notably, the Support Vector Regressor underperformed significantly, suggesting it is not well-suited for this dataset.

To qualitatively evaluate prediction accuracy, we analyzed model output for several specific LEGO sets using our best model – the De-

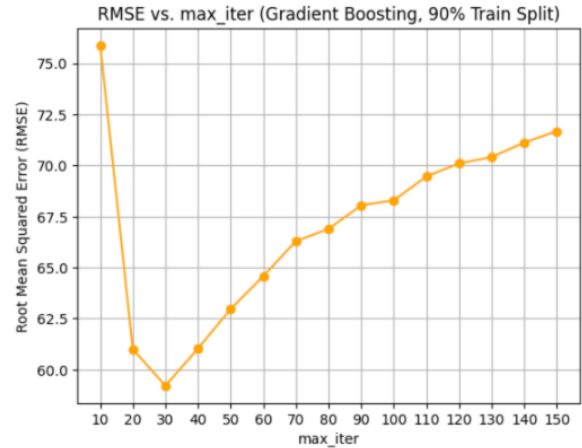


Figure 13: *Gradient Boosting RMSE vs. Iteration Count*

Model	R^2	MAE	RMSE
Mean Baseline	-0.001	72.914	115.637
Median Baseline	-0.199	72.975	126.575
Gaussian Random Baseline	-1.508	140.425	183.018
Linear Regression	0.7005	41.3295	72.4270
Decision Tree (Depth = 7)	0.8011	33.1857	59.0294
Random Forest (Depth = 11)	0.7740	30.3675	64.9245
Gradient Boosting (iter = 30)	0.7999	32.7536	59.2072
Support Vector Regressor	0.4794	46.9756	95.5005

Table 7: *Best Results Across Models (MAE, RMSE, R^2)*

cision Tree Regressor with max depth 7 and a 90-10% train-test split. Table 8 compares predicted and actual prices across ten sets. For most sets, including mid-range and low-value ones, the predicted price closely approximated the true resale value. However, large discrepancies emerged with extremely high-priced sets like the Titanic, which had a predicted price of \$456 and a true price of \$640.

This suggests that while our model captures typical pricing patterns well, it still struggles with high-end outliers. Future work may involve integrating external data, such as popularity trends or historical price trajectories, to improve accuracy on these edge cases. Together, these experimental results validate

	Predicted Price	Actual Price	Set Name
0	456.887	640.000	Titanic
1	27.268	10.000	Drone Explorer
2	93.962	60.133	Clay’s Falcon Fighter Blaster
3	29.904	22.250	Magical Funfair Stalls
4	122.247	124.990	Poe’s X-Wing Fighter
5	14.530	15.000	Mini Trains
6	93.962	190.000	Dragon Mountain
7	95.268	64.990	Organic Grocery Store
8	27.268	24.990	Wolf Legend Beast
9	98.181	80.000	Container Heist

Table 8: *Predicted vs Actual Prices (Best Model: Decision Tree, Depth = 7)*

the utility of machine learning for predicting LEGO resale prices and highlight the Decision Tree model as a strong candidate for further refinement.

8 Conclusions/Future Work

In this project, we developed a machine learning pipeline to predict the resale price of LEGO sets using structured attributes such as theme, year, number of pieces, and original MSRP. Our work is situated at the intersection of data science and alternative investments, where collectible assets like LEGO have demonstrated strong appreciation potential and considerable price volatility. Starting with a public dataset of nearly 15,000 entries, we filtered and cleaned the data to retain 3,607 high-quality entries. We then engineered relevant features, applied one-hot encoding for categorical data, and evaluated the predictive performance of multiple regression models.

Across our modeling experiments, we tested five machine learning algorithms – Linear Regression, Decision Tree Regressor, Random Forest Regressor, Gradient Boosting Regressor, and Support Vector Regressor – alongside three non-learning baselines. We used hold-out validation with varying train-test splits and tuned critical hyperparameters for the best-performing models. Among all, the De-

cision Tree Regressor with a maximum depth of 7 achieved the best results, with an R^2 of 0.8011 and an RMSE of 59.0294. This demonstrated not only the predictability of LEGO resale prices but also the value of non-linear models in capturing complex interactions among product features. Theme, despite being a categorical variable with low direct correlation to price, proved essential for accurate predictions, especially when encoded properly.

Our experiments surfaced several important takeaways. First, structured attributes are highly informative, and resale prices can be effectively modeled using historical data. Second, tree-based models consistently outperformed both linear models and trivial baselines, particularly after hyperparameter tuning. Third, baseline models provided a clear performance floor, confirming that our machine learning models were indeed learning meaningful relationships. Fourth, while numeric features like MSRP and piece count were important, theme-level categorization contributed significantly to model accuracy. However, the model did struggle with high-end outliers such as the Titanic set, suggesting that structured metadata alone may be insufficient to fully explain extreme pricing behavior influenced by factors like hype, scarcity, or cultural resonance.

Looking forward, we plan to enhance the model by incorporating external data sources. This could include scraping historical price trends from platforms like eBay, leveraging Google Trends for cultural relevance signals, and applying natural language processing (NLP) to extract insights from LEGO set names, such as identifying terms like “Collector’s Edition” or “Exclusive.” We also aim to integrate interpretability techniques like SHAP values to provide deeper transparency into how feature inputs drive predictions. Further, we may explore the application of neural networks or time-aware models to account for longer-term market dynamics.

Lastly, the methods developed in this study offer strong potential for generalization to other collectible domains. Our modeling framework could be extended to predict the resale value

of trading cards (e.g., Pokémon, Magic: The Gathering, Yu-Gi-Oh!), vintage video games (e.g., NES, Sega Genesis, PlayStation), and collectible toys and action figures (e.g., Funko Pops, Hot Wheels, Star Wars figures). These domains share key structural similarities with LEGO sets, including categorical identifiers, condition ratings, release years, and high-value secondary markets. As such, our approach lays the groundwork for scalable, interpretable prediction tools across a range of alternative asset classes – empowering collectors, resellers, and investors to make data-informed decisions in dynamic, speculative markets.

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